MEASURING THE MOAT: EXCESS BOUNDS NEAR SEYMOUR'S SECOND NEIGHBOURHOOD CONJECTURE

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ABSTRACT. Seymour's second neighbourhood conjecture (SSNC) asserts that every oriented graph has a vertex whose second out-neighbourhood is at least as large as its first. For an oriented graph G define $\mathrm{exc}(G) = \sum_v \max(0,\, d_2(v) - d_1(v) + 1)$; then $\mathrm{exc}(G) = 0$ iff G is a counterexample, and SSNC is the statement $\mathrm{exc} \geq 1$. Since excess 1 is trivially attained using sinks, we study $E(n)$, the minimum excess over strongly connected oriented graphs on n vertices with minimum out-degree at least 8 — the bound forced on counterexamples by Kaneko–Locke and the recent $\delta^+ = 7$ elimination.

We prove the constructive upper bounds $E(mt) \leq m$ for $m \geq 3$, $t \geq 8$, extend them to cycle-power skeletons, and obtain $E(n) \leq 5$ at the primes $n = 47, 53, 59$ by transplanting a locally rigid structure discovered by stochastic search. We report machine-verified upper bounds at selected orders $24 \leq n \leq 75$; all witnesses are published with hashes and re-verifiable by three independent implementations. A pre-registered negative control documents that search attainment values are not floors. A computer-assisted infeasibility computation gives $E(17) \geq 3$, and the best value observed at $n = 50$ remains unchanged in our searches when the degree bound is raised to $\delta^+ \geq 9, 10$. Analysis of the best-known witnesses reveals two distinct local architectures with identical local rigidity. We conjecture a uniform lower bound $E(n) \geq 3$ and pose the regularity of $n \mapsto E(n)$ as an open problem.

1. INTRODUCTION

Seymour's second neighbourhood conjecture [DL95] states:

Conjecture 1.1 (SSNC)**.** *Every oriented graph (digraph without 2-cycles) contains a vertex v with $|N_2^+(v)| \geq |N_1^+(v)|$.*

The conjecture is proved for tournaments [Fis96, HT00] and in many special classes; every vertex satisfies $|N_2^+(v)| \geq \gamma\,|N_1^+(v)|$ for some vertex with $\gamma = 0.715538\ldots$ [HP24], improving [CSY03]. A counterexample must have minimum out-degree $\delta^+ \geq 8$: out-degree ≤ 6 was eliminated by [KL01] and $\delta^+ = 7$ by [SSS26] via a signature-compression argument checked by CP-SAT.

Recent work has begun to study the *equality boundary* of the conjecture. [Hal26] calls a strongly connected oriented graph **Pisa** if $\max_v(|N_2^+(v)| -$

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$|N_1^+(v)|) = 0$, i.e. no vertex exceeds and at least one vertex meets the bound; [GKZ26] study **Seymour-tight** orientations, in which *every* vertex meets the bound, and develop a product calculus and an algebraic classification for them. In the tight case the set of boundary vertices is everything; in a generic Pisa graph it can be much smaller. This note measures how small.

Definitions. For an oriented graph G write $d_1(v) = |N_1^+(v)|$ and $d_2(v) = |N_2^+(v)|$, where $N_2^+(v) = \{x : \mathrm{dist}^+(v, x) = 2\}$; in particular $N_1^+(v)$ and $N_2^+(v)$ are disjoint. Call $m(v) = d_2(v) - d_1(v)$ the *margin* of v. Call v **tight** if $m(v) = 0$ and **satisfied** if $m(v) \geq 0$ (a "Seymour vertex"). Define

$$\mathrm{exc}(G) \;=\; \sum_{v \in V(G)} \max\bigl(0,\; m(v) + 1\bigr).$$

Then $\mathrm{exc}(G) = 0$ iff G is a counterexample to SSNC; SSNC is the statement $\mathrm{exc} \geq 1$; and for a Pisa graph the excess equals the number of tight vertices. Adding a universal sink produces $\mathrm{exc} = 1$ trivially [GKZ26, §1], so the interesting quantity requires the degree bound that counterexamples must satisfy:

$$E(n) \;=\; \min\{\,\mathrm{exc}(G) : |V(G)| = n,\; G \text{ strongly connected oriented},\; \delta^+(G) \geq 8\,\}.$$

(We deliberately restrict to strongly connected graphs, motivated by the minimal-counterexample reduction — a closed out-section of a counterexample is a smaller counterexample, cf. [EDGGK25]; the unrestricted quantity can differ (sinks give excess 1) and is not studied here.)

$E(n) = 0$ for some n iff SSNC is false. If SSNC is true, then $E(n) \geq 1$ for all n, and any stronger uniform lower bound is a strengthening of the conjecture. The contributions of this note:

(1) **Upper bounds by construction** (§2): a single "out-set inclusion" lemma yields both a family of *pure-ring* constructions ($E(mt) \leq m$ for $m \geq 3$, $t \geq 8$; generalized to cycle-power skeletons) and, after transplant of a search-discovered local structure, the bound $E(n) \leq 5$ at the primes $n = 47, 53, 59$.

(2) **Measurements** (§3): a table of best known upper bounds at selected orders between 24 and 75 with published, hash-identified witnesses; and a measurement-theoretic caveat we learned the hard way — *search attainment is not the floor* — documented via a controlled experiment.

(3) **Lower bounds and rigidity** (§4): a computer-assisted infeasibility computation gives $E(17) \geq 3$; invariance of the best value observed at $n = 50$ under $\delta^+ \geq 9, 10$; and twelve one-hour UNKNOWNs for the tournament-restricted question $\mathrm{exc} \leq 2$, $17 \leq n \leq 28$.

(4) **Structure of best-known witnesses** (§5): two architectures, one dynamics.

(5) **Open problems** (§6), foremost: does a uniform constant lower bound $E(n) \geq c \geq 2$ hold, and what governs the apparent floor $3 \leq E(n) \leq 5$?

Verification protocol. Every witness in this note is stored in the accompanying repository as an adjacency list with a SHA-1 fingerprint, and every numerical claim is checked by three implementations written independently (a numpy pipeline used by the search; a pure-Python verifier; and a standard-library-only verifier written from the definitions without access to the other two). A `make verify-*` target reproduces each claim.

2. Upper bounds by construction

2.1. **The out-set inclusion lemma.** All constructions below are perturbed blow-ups of directed cycles. The mechanism that keeps perturbations from leaking is the following observation.

Lemma 2.1 (out-set inclusion)**.** *Let G be an oriented graph and let $v \neq p$ be non-adjacent vertices with $N_1^+(p) \subseteq N_1^+(v)$. Let $G' = G + (v \to p)$. Then G' is oriented, and:*

(i) *if $p \in N_2^+(G, v)$, then $m_{G'}(v) = m_G(v) - 2$;*
(ii) *if $p \notin N_1^+(G, v) \cup N_2^+(G, v)$, then $m_{G'}(v) = m_G(v) - 1$;*
(iii) *for $x \neq v$, margins change only by the possible appearance of p in $N_2^+(x)$ through the new path $x \to v \to p$: precisely, $m_{G'}(x) = m_G(x) + 1$ exactly when $v \in N_1^+(x)$ and $p \notin N_1^+(x) \cup \{x\} \cup N_2^+(G, x)$; all other margins are unchanged. In particular, if every in-neighbour x of v satisfies $p \in N_1^+(x)$, then no margin other than $m(v)$ changes.*

Proof. G' is oriented since $p \to v$ is absent by non-adjacency. For v: $d_1(v)$ increases by exactly one. Candidate new second neighbours of v are the out-neighbours of the new first neighbour p, i.e. $N_1^+(p) \setminus (N_1^+(G', v) \cup \{v\}) = \emptyset$ by hypothesis; and no old second neighbour $u \neq p$ is lost, since arcs are only added and $u \notin N_1^+(G', v)$. The only possible change to $N_2^+(v)$ is the removal of p itself, which occurs precisely in case (i); this gives (i) and (ii). For $x \notin \{v, p\}$: $d_1(x)$ is unchanged, and the only new directed paths of length two are of the form $x \to v \to p$, which affect membership of p alone; the stated condition characterizes when p actually enters $N_2^+(x)$. For $x = p$: no new path leaves p, and no new path of length two into $N_2^+(p)$ is created, since such a path would require an arc into v to be new. □

In words: *if the target's out-set is already inside the shooter's first neighbourhood, the shot costs the shooter one unit of margin and is invisible to bystanders who see both.* The same arithmetic underlies the in-star trick in the generalized lexicographic products of [GKZ26] — and, as §5 shows, it was re-invented independently by stochastic search.

2.2. **Pure rings.**

Theorem 2.2. *Let $m \geq 3$ and $t \geq 8$, and let $n = mt$. Then $E(n) \leq m$.*

Construction. Partition V into layers $L_0, \dots, L_{m-1}$ of size t; add all arcs $L_i \to L_{i+1}$ (indices mod m); in each layer choose one *pure* vertex p_i and add the arc $v \to p_i$ for every impure $v \in L_i \setminus \{p_i\}$. Every impure vertex has $d_1 = t + 1,\ d_2 = t$, hence margin -1; every pure vertex has $d_1 = d_2 = t$, margin 0. The graph is digon-free (arcs between layers all point forward; inside a layer only $\to p_i$ arcs exist and p_i sends none), strongly connected, with $\delta^+ = t \geq 8$, and $\mathrm{exc} = m$. In the layered construction the target p_i lies in the shooter's own layer, at directed distance $m \geq 3$ from every impure $v \in L_i$ in the unperturbed layered graph, so case (ii) of Lemma 2.1 applies and each intra-layer arc costs its shooter exactly one unit of margin; every in-neighbour of v (all of L_{i-1}) sees p_i in its first neighbourhood, so by (iii) no other margin moves. $\square$

Theorem 2.3 (cycle-power skeletons)**.** *Let m, k with $2k < m$, and $t \geq 1$ with $kt \geq 8$; let $n = mt$. Then $E(n) \leq m$.*

Construction. Replace the cycle skeleton by the k-th power $\vec{C}_m^k$: each layer sends complete arcs to the next k layers. Purity and the in-star are as before; the out-set inclusion is now even cleaner, since the target's out-set lies entirely inside the shooter's first neighbourhood. In the skeleton $\vec{C}_m^k$ the shortest directed path from an impure $v \in L_i$ back to its own layer has length at least $\lceil m/k \rceil \geq 3$ (since $m > 2k$), so case (ii) of Lemma 2.1 applies exactly as before; the in-neighbour condition of (iii) holds as every layer L_{i-j} $(1 \leq j \leq k)$ sees p_i in its first neighbourhood. Pure vertices have $d_1 = d_2 = kt$; impure vertices have $d_1 = kt + 1$, $d_2 = kt$. This covers e.g. $n = 25$ (m=5, k=2, t=5), $n = 35$ $(5, 2, 7)$, $n = 49$ $(7, 2, 7)$. $\square$

Uneven layer sizes generally fail; the algebraic reason is the kernel condition for generalized lexicographic products identified in [GKZ26, Thm 4.6], which for the plain cycle forces equal sizes. Our constructions are *not* Seymour-tight — they sit at the opposite extreme of the Pisa class, with as few tight vertices as the skeleton allows.

2.3. **Keystone transplants: upper bounds at primes.** Ring constructions require $m \mid n$; at e.g. $n = 47, 53, 59$ no cycle-power skeleton from Theorem 2.3 yields a bound comparable to 5 (the theorem applies there only vacuously, with $m = n$, $t = 1$, giving $E(n) \leq n$). Stochastic search (§3) found a locally rigid record witness at $n = 53$ with a different local architecture: a **keystone cluster** — a satisfied vertex u of margin $+1$ together with *wing* vertices w whose out-sets are exact copies of $N_1^+(u)$ plus the single arc $w \to u$ (verified equality of out-sets in the witness; see `data/cluster_structure_n53.txt`). This is the out-set inclusion mechanism again, asymmetric and ring-free.

Guided by this structure, a deterministic transplant procedure (delete or clone bulk vertices while preserving the cluster; `constructions/transplant.py`) produced explicit witnesses establishing

$$E(47) \leq 5, \qquad E(53) \leq 5, \qquad E(59) \leq 5,$$

each verified by the three-implementation protocol, and exc $= 4$ at $n = 57$ with *no* $+1$ vertex (all four survivors tight) — showing the $+1$ "insurance" is not intrinsic to the family (at $n = 57$ the ring bound $E \leq 3$ is anyway better). We emphasise that these are individually verified computational objects: we do not currently have a general transplant lemma giving conditions under which the cluster embeds in a host of arbitrary order with controlled excess (Problem 6 in §6).

3. Measurements

3.1. **Table of best known upper bounds.** The full table is machine-generated from the repository data (`note/make_table.py`); hashes are canonical-adjacency SHA-1 prefixes from the witness manifest.

n	construction	search	best known	witness
24	3 (ring m=3,k=1,t=8)	-	3	`2517130b`
25	5 (ring m=5,k=2,t=5)	12	5	`93072b44`
27	3 (ring m=3,k=1,t=9)	-	3	`5b5ae497`
30	3 (ring m=3,k=1,t=10)	3	3	*(thm)*
35	5 (ring m=5,k=2,t=7)	7	5	`5a88d73d`
40	4 (ring m=4,k=1,t=10)	23	4	*(thm)*
45	3 (ring m=3,k=1,t=15)	21	3	*(thm)*
47	5 (keystone transplant)	9	5	`21f36278`
48	3 (ring m=3,k=1,t=16)	13 (seed-free)	3	`d43ca1db`
49	7 (ring m=7,k=2,t=7)	6	6	`ead3c3c3`
50	5 (ring m=5,k=1,t=10)	5 (20 seed-free)	5	`17cd7dc8`
51	3 (ring m=3,k=1,t=17)	-	3	`e5d2d5eb`
53	5 (keystone transplant)	5	5	`dc922e89`
57	3 (ring m=3,k=1,t=19)	-	3	`8210dcab`
59	5 (keystone transplant)	7	5	`3eedc493`
60	3 (ring m=3,k=1,t=20)	33	3	*(thm)*
75	3 (ring m=3,k=1,t=25)	3	3	*(thm)*

Search values marked "seed-free" are the negative controls of §3.2. At $n = 57$ the transplant family also realises exc $= 4$ (all survivors tight), which is not a bound record there.

3.2. **Search attainment is not the floor.** We initially read fresh-search values at primes (47:9, 59:7, 53:5) as a "prime staircase". A pre-registered controlled experiment (fresh search on the *composites* $n = 48, 50$ with all construction seeds withheld) returned 13 and 20 — far above the construction bounds 3 and 5 at the same n. Conclusion: **fresh-search values measure search reachability, not the floor**; the staircase was a measurement artefact, later confirmed when the keystone transplant put all three primes at 5. We therefore report construction bounds and search attainments in separate columns throughout. (Full logs, pre-registration commits, and the negative-control seed runs are in the repository.)

3.3. **Observed δ-invariance.** At $n = 50$ the best value observed, 5, is unchanged when the degree bound is raised to $\delta^+ \geq 9$ and $\delta^+ \geq 10$ (the divisor-ring witness has $\delta^+ = 10$ and satisfies all three constraints; forty generations of search under each constraint produced no improvement). Any conjectured lower bound may thus plausibly be sought in a form not depending on δ beyond the counterexample threshold.

4. Lower bounds and rigidity

Proposition 4.1 (computer-assisted)**.** *Every oriented graph on 17 vertices with $\delta^+ \geq 8$ satisfies* exc ≥ 3 *(in particular $E(17) \geq 3$).*

Proof. By the encoding of Appendix A, an oriented graph on 17 vertices with $\delta^+ \geq 8$ and exc ≤ 2 exists only if the CP-SAT model $M_{17,2}$ is feasible (no strong-connectivity constraint is needed or used; see Appendix A). OR-Tools 9.15.6755 reports INFEASIBLE (357.0 s, fixed seed, 8 workers; model hash `1d22eb15…`, record in `data/excess2_results.jsonl`). The conclusion is therefore conditional on the correctness of the encoding (argued in Appendix A) and the soundness of the solver; no independently checkable proof certificate is currently produced, and obtaining one via a proof-logging solver is left as future work. □

For $18 \leq n \leq 22$ the same computation is UNKNOWN within 3600 s each; the direct method's reach ends near $n = 17$.

Tournament rigidity. By [HT00], sink-free tournaments have at least two satisfied vertices; whether *exactly two, both tight* is attainable in a tournament (which would give exc $= 2$ and refute any $E \geq 3$-type bound in the tournament-dense regime) appears to be open. Our CP-SAT scans of the tournament-restricted question exc ≤ 2 for $17 \leq n \leq 28$ returned twelve consecutive UNKNOWNs at one hour each: the question is not cheaply decidable in either direction. We record this as an explicit open problem (§6).

5. Structure of best-known witnesses: two architectures, one dynamics

Two best-known witnesses at exc $= 5$:

- $n = 50$ (`17cd7dc8`): five tight survivors forming a **pentagon ring** $8 \to 16 \to 23 \to 36 \to 47 \to 8$ in which each survivor's first neighbourhood contains the next survivor and its second neighbourhood the one after; the sensitivity matrix (all valid single arc-flips) is diagonal; for four of the five survivors the cheapest kill is an exact equal trade ($\Delta E = 0$, a reserve vertex rising from margin -1), while the fifth (v_{36}) costs $\Delta E = +2$.

- $n = 53$ (`dc922e89`): a **keystone cluster** — margins $(+1, 0, 0, 0)$ on $\{18; 34, 35, 36\}$ with the wings' out-sets equal to the keystone's; the sensitivity matrix shows the only non-diagonal row at the keystone: killing it raises both wings.

Different architecture and different economy (the ring witness's deficits are shallow (margin histogram $\{-5{:}1,\ -4{:}1,\ -3{:}4,\ -2{:}17,\ -1{:}22\}$), while the keystone witness carries a deep deficit tail reaching -15) — yet identical dynamics: both admit no improving single-flip move and lie on large neutral plateaus, and the keystone's apparent $+1$ slack survives an exhaustive two-flip attack restricted to the cluster neighbourhood: all 171,288 valid coordinated pairs (f_1, f_2), where f_1 ranges over arc modifications (addition, deletion, reversal, preserving orientation and $\delta^+ \geq 8$) incident to the keystone vertex and f_2 over such modifications incident to the 20-vertex cluster closure, with validity of f_2 re-evaluated after applying f_1. No pair decreases the excess; 7,712 pairs are neutral. The corresponding exhaustive depth-2 search has not been performed for the $n = 50$ witness. The local rigidity of both witnesses is insensitive to the architecture, the economy, and (§3.3) the degree bound — consistent with, though not implying, a global counting obstruction rather than a local structural one.

Remark 5.1 (Pisa structure). All witnesses above are Pisa graphs whose underlying undirected graphs are irregular and far from K_n-minus-a-matching; together with small blow-ups of directed cycles (e.g. the 2-blow-up of $\vec{C}_4$ on 8 vertices) they show that the classification of underlying graphs of Pisa graphs is substantially richer than conjectured in [Hal26, Conj. 5.1]. We have communicated explicit counterexamples to the author (private communication, July 2026).

6. Open problems

(1) **Uniform floor.** Is there $c \geq 2$ with $E(n) \geq c$ for all n? All evidence above is consistent with $E(n) \geq 3$; we know only $E(17) \geq 3$ and, conditionally on SSNC, $E(n) \geq 1$.
(2) **Regularity of E.** The best known upper bounds lie in $\{3, 4, 5, 6\}$ among the measured orders between 24 and 75, with the value governed by available skeletons, not by primality. Is E eventually constant? Monotone in any sense?
(3) **Tournaments with exactly two tight vertices.** Does a sink-free tournament with exactly two satisfied vertices, both tight, exist? (Twelve one-hour CP-SAT attempts, $17 \leq n \leq 28$, all UNKNOWN.)
(4) **Keystone family limits.** The transplant family attains 4 (at $n = 57$) with all survivors tight; is 3 attainable ring-free? Is the family's native floor below the ring floor anywhere?

(5) **Global counting.** Find a double-counting / matrix-analytic proof of any bound $E(n) \geq 2$. The matrix S_D of [GKZ26] expresses margins as row sums; excess is the positive part of $S_D\mathbf{1}$. The spectral theory of S_D (untouched in the literature) and LP duality over flow relaxations both suggest themselves.

(6) **Transplant lemma.** Formulate and prove a general lemma specifying when the keystone cluster of §2.3 can be embedded in a host graph of given order with excess bounded by a constant, turning the transplant procedure from a witness generator into a theorem.

7. Methods and reproducibility

Search: island-model program evolution (LLM mutation operators acting on construction programs), with every discovered improvement persisted to disk at discovery time under a SHA-1 fingerprint. Verification: three independent implementations (§1); every claim maps to a `make verify-*` target. Pre-registration: search predictions were committed to the repository before the corresponding runs (commit hashes in `docs/`). Division of labour: direction and decisions by the human author; supervision, literature, and independent verification by Claude (Fable); implementation, search, and solver engineering by Claude (Code). All witness-based numerical claims are independently machine-verifiable from the accompanying repository; the computer-assisted infeasibility result has the limitations stated in Appendix A.

Total LLM API cost of all search and verification reported here: USD 65.54.

Repository: [URL on publication; private at draft time].

Appendix A. The CP-SAT model $M_{n,c}$

We describe the model behind Proposition 4.1 completely, so that it can be checked against the paper alone; the implementation is `experiments/delta8/excess2_search.py` (model hash and solver version are recorded with the run).

Variables. For ordered pairs $i \neq j$ of $[n]$, Boolean *arc* variables a_{ij}; for ordered pairs $v \neq w$, Boolean *distance-two* variables s_{vw}; for each v, an integer $e_v \in \{0, \ldots, n\}$.

Constraints.

(O) $a_{ij} + a_{ji} \leq 1$ for all $i < j$ (orientation: no digons; loops are excluded syntactically);

(D) $\sum_{j \neq i} a_{ij} \geq 8$ for all i (minimum out-degree);

(S1) $s_{vw} + a_{vw} \leq 1$ (N_1^+ and N_2^+ disjoint);

(S2) $a_{vy} + a_{yw} - a_{vw} - s_{vw} \leq 1$ for all distinct v, y, w (if a two-path $v \to y \to w$ exists and $a_{vw} = 0$, then $s_{vw} = 1$);

(E) $e_v \geq \sum_{w \neq v} s_{vw} - \sum_{j \neq v} a_{vj} + 1$ (per-vertex excess lower bound; $e_v \geq 0$ by its domain);

(C) $\sum_v e_v \leq c$;

(B) symmetry breaking: $\sum_j a_{0j} \ge \sum_j a_{ij}$ for all i, and $a_{0,j} \ge a_{0,j+1}$ for $1 \le j \le n-2$ (vertex 0 has maximum out-degree and its out-neighbourhood is an initial segment).

The graph-to-model direction.

Lemma A.1 (graph-to-model direction)**.** *If an oriented graph G on n vertices with $\delta^+(G) \ge 8$ and $\operatorname{exc}(G) \le c$ exists, then $M_{n,c}$ is feasible.*

Proof. Relabel G so that vertex 0 has maximum out-degree and its out-neighbours are $1, \dots, d_1(0)$; this meets (B). Set $a_{ij} = 1$ iff $i \to j$ in G; (O) holds since G is oriented and (D) since $\delta^+ \ge 8$. Set $s_{vw} = 1$ iff $\operatorname{dist}^+(v, w) = 2$; (S1) holds by disjointness of the neighbourhoods, and (S2) holds because a two-path with $a_{vw} = 0$ witnesses distance exactly two. Then $\sum_w s_{vw} = d_2(v)$, so setting $e_v = \max(0, d_2(v) - d_1(v) + 1)$ satisfies (E), and (C) is $\operatorname{exc}(G) \le c$. □

Hence INFEASIBLE implies that no such graph exists, proving Proposition 4.1.

Strong connectivity is not encoded. The model deliberately contains **no** strong-connectivity constraint, and Lemma A.1 correspondingly assumes none: *every* oriented graph with $\delta^+ \ge 8$ and $\operatorname{exc} \le c$, strongly connected or not, yields a feasible assignment. Consequently INFEASIBLE rules out the *larger* class, which is why Proposition 4.1 is stated for all oriented graphs and the bound $E(17) \ge 3$ for the strongly connected subclass follows a fortiori. Omitting a constraint is a relaxation; a relaxation can only turn infeasible instances feasible, never the reverse, so in the infeasibility direction it preserves — indeed strengthens — the conclusion. The same encoding (without strong connectivity) underlies the budget-exhausted scans reported in §4 ($18 \le n \le 22$ and the tournament-restricted question); those runs return UNKNOWN and therefore claim nothing in either direction.

Feasibility direction. The second deliberate relaxation: s_{vw} may be 1 without a witnessing two-path (it is forced upward by (S2), never downward), so a *feasible* assignment need not describe a real graph of low excess. For this reason every FEASIBLE output of this model elsewhere in the project is re-verified exactly on the extracted adjacency matrix before being believed; INFEASIBLE outputs need no such step.

Limitations. The three-implementation protocol of §1 verifies *witnesses*; it does not independently verify *infeasibility*, which rests on the encoding above and on the soundness of the CP-SAT solver. Producing an independently checkable certificate (e.g. via a proof-logging SAT/PB solver) is future work.

References

[CSY03] Guantao Chen, Jian Shen, and Raphael Yuster, *Second neighborhood via first neighborhood in digraphs*, Annals of Combinatorics **7** (2003), 15–20.

[DL95] Nathaniel Dean and Brenda J. Latka, *Squaring the tournament—an open problem*, Congressus Numerantium **109** (1995), 73–80.

[EDGGK25] Alberto Espuny Díaz, António Girão, Bertille Granet, and Gal Kronenberg, *Seymour's second neighbourhood conjecture: random graphs and reductions*, Random Structures & Algorithms **66** (2025), no. 1, e21251, arXiv:2403.02842.

[Fis96] David C. Fisher, *Squaring a tournament: a proof of Dean's conjecture*, Journal of Graph Theory **23** (1996), 43–48.

[GKZ26] Krystal Guo, Ross J. Kang, and Gabriëlle Zwaneveld, *Seymour-tight orientations*, 2026, arXiv:2603.29626.

[Hal26] Stanisław M. S. Halkiewicz, *Extremal instances of Seymour's second neighbourhood conjecture*, 2026, arXiv:2601.21563; title of v3 (May 2026), earlier versions under a different title.

[HP24] Hao Huang and Fei Peng, *An improved bound on Seymour's second neighborhood conjecture*, 2024, arXiv:2412.20234.

[HT00] Frédéric Havet and Stéphan Thomassé, *Median orders of tournaments: a tool for the second neighborhood problem and Sumner's conjecture*, Journal of Graph Theory **35** (2000), 244–256.

[KL01] Yoshihiro Kaneko and Stephen C. Locke, *The minimum degree approach for Paul Seymour's distance 2 conjecture*, Congressus Numerantium **148** (2001), 201–206.

[SSS26] Arpan Sadhukhan, R. B. Sandeep, and Sagnik Sen, *A proof of Seymour's second neighborhood conjecture for oriented graphs with minimum out-degree equal to 7*, 2026, arXiv:2606.30588.

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